Assignment - I

1) Let $A \in F^{n \times n}$. Show that a scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A iff the map $A - \lambda I : F^{n \times 1} \to F^{n \times 1}$ is not one to one.

2) Let $A \in F^{n \times n}$. Show that A is invertiable iff 0 is not an eigenvalue of A.

3) Find all eigenvalues and corresponding eigenvectors of the following matrices:

a) $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$

4) Find all eigenvalues and their corresponding eigenvectors of the $n \times n$ matrix A whose

j^{th} row has each entry j.

5) Let $p(t) = t^3 - \alpha t^2 - (\alpha+3)t - 1$ for a real number α. Consider

$$A = \begin{bmatrix} \alpha & \alpha+3 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P = \begin{bmatrix} -1 & 2 & -\alpha-3 \\ 1 & -1 & \alpha+2 \\ -1 & 1 & -\alpha-1 \end{bmatrix}$$

a) Compute the characteristic polynomials of A and $P^{-1}AP$.

b) Show that all zeros of $p(t)$ are real regardless of the value of α.

6) Let A be a 4×4 non-invertible matrix with all diagonal entries 1. If $2+3i$ is an eigenvalue of A what are its other eigenvalues?

7) Construct a 3×3 hermitian matrix with no zero entries whose eigenvalues are 1, 2 and 3.

8) Construct a 2x2-real skew symmetric matrix whose eigenvalues are purely imaginary.

9) Construct an orthogonal 2x2-matrix whose determinant is 1.

10) Construct an orthogonal 2x2-matrix whose determinant is -1.

11) Let $a+ib$ be an eigenvalue of $A \in R^{n \times n}$ with an associated eigenvector $x+iy$,

where $a, b \in \mathbb{R}$, $x, y \in \mathbb{R}^{n\times 1}$.

Show the following.

a) The vectors $x+iy$ and $x-iy$ are linearly independent in $\mathbb{C}^{n\times 1}$.

b) The vectors x and y are linearly independent in $\mathbb{R}^{n\times 1}$.

12) An $n\times n$-matrix A is said to be idempotent if $A^2 = A$. Show that the only possible

eigenvalues of an idempotent matrix are 0 or 1.

(13) An $n \times n$-matrix A is said to be nilpotent if $A^k = 0$ for some k.

Show that 0 is the only eigenvalue of a nilpotent matrix.

(14) Let $A = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$,

$B = \begin{bmatrix} -1 & 3 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 3 \\ 0 & -1 \end{bmatrix}$.

Show that both B and C are Schur triangularizations of A.

15) Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 7 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 2 \\ 0 & 7 \end{bmatrix}$. Show that B is a Schur triangularization of A, but C is not.

16) Determine a matrix A such that $A^*A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & i \\ 0 & -i & 1 \end{bmatrix}$

17) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

Compute A^{-1} and A^{10}.

18) Let $A \in F^{n \times n}$. Let $p(t)$ be a monic polynomial of degree n. If $p(A) = 0$, then does it follow that $p(t)$ is the characteristic polynomial of A?

19) Diagonalize the given matrix, and then compute its fifth power:

a) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ b) $\begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$

c) $\begin{bmatrix} 7 & -5 & 15 \\ 6 & -4 & 15 \\ 0 & 0 & 1 \end{bmatrix}$

20) Show that the following matrices are diagonalized by

matrices in $\mathbb{R}^{3\times 3}$.

a) $\begin{bmatrix} 3/2 & -1/2 & 0 \\ -1/2 & 3/2 & 0 \\ 1/2 & -1/2 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$

21) Check whether each of the following matrices is diagonalizable. If diagonalizable, find a

basis of eigenvectors for $\mathbb{C}^{3\times 1}$.

a) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$ b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

22) Find orthogonal or Unitary diagonalizing matrices for the following:

a) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ b) $\begin{bmatrix} 1 & 3+i \\ 3-i & 4 \end{bmatrix}$

c) $\begin{bmatrix} -4 & -2 & 2 \\ -2 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

23) Determine A^5 and a matrix B such that $B^2 = A$, where

a) $A = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$

b) $A = \begin{bmatrix} 9 & -5 & 3 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}$